

**BANARAS HINDU UNIVERSITY**

B.Voc. Semester IV Examination, 2022-23

**Paper: GEN-402 — Data Structures Algorithms**

Time: Three hours

Maximum Marks: 70

*Instructions: Attempt all questions. Symbols have their usual meanings.***SECTION — A**

Long-Answer Questions. Answer any two questions:

**2 × 12 = 24 Marks**

**Question 1.** Define Linked Lists. What are the basic operations performed on a Linked List? Write methods for the creation of a Linked List and insertion of any three elements at the beginning of the list.

**Question 2.** What is a Binary Tree? Explain Preorder, Inorder, and Postorder traversals of a binary tree with suitable examples.

**Question 3.** What is a Graph data structure? Explain the Breadth First Search (BFS) algorithm with a suitable example. Differentiate between Breadth First Search and Depth First Search.

**Question 4.** Explain Stack Data Structure. Write a program to implement a stack using an array (show creation, insertion, deletion, and display of elements).

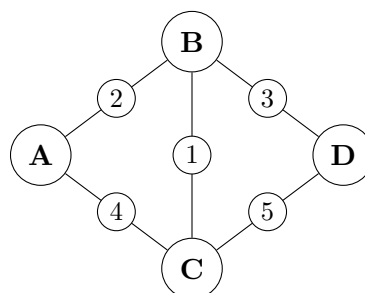
**SECTION — B**

Attempt any six questions:

**6 × 6 = 36 Marks**

**Question 5.** Explain Binary Search with a suitable example. What is the prerequisite condition to implement binary search?

**Question 6.** Explain Prim's algorithm to find a Minimum Spanning Tree (MST). Find the cost of the minimum spanning tree of the following graph:



**Question 7.** Explain Bubble Sort with a suitable example.

**Question 8.** What is recursion? Write a program to find the factorial of a given number using recursion.

**Question 9.** Explain Insertion Sort using a suitable example.

**Question 10.** Give the algorithm to evaluate a postfix expression using a stack. Evaluate the postfix expression  $5\ 2\ 4\ * + 7\ -$  using the algorithm and show the stack contents at each step.

**Question 11.** What is an AVL tree? Explain LL and LR rotations.

**Question 12.** Define a queue data structure. Explain the operations enqueue and dequeue in a queue.

**Question 13.** Write a Java program to find the minimum and maximum elements in an array of 20 elements. (The elements must be input by the user).

## SECTION — C

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**Question 14.** Attempt all parts:

**10 × 1 = 10 Marks**

(a) A Binary Search Tree (BST) has the property that the value of each node in the left subtree is \_\_\_\_\_ the value of the root node, and the value of each node in the right subtree is \_\_\_\_\_ the value of the root node.

- (a) Greater, Greater
- (b) Greater, Less
- (c) Less, Less
- (d) Less, Greater

(b) In a stack, which operation is used to remove an element from the stack?

- (a) Pop
- (b) Push
- (c) Peek
- (d) Empty

(c) Which of the following data structures is not a linear data structure?

- (a) Stack
- (b) Queue
- (c) Linked List
- (d) Binary Tree

(d) Which of the following data structures finds its use in recursion?

- (a) Stack

- (b) Array
  - (c) Linked List
  - (d) Queue
- (e) Which of the following is a Divide and Conquer algorithm?
- (a) Bubble Sort
  - (b) Selection Sort
  - (c) Heap Sort
  - (d) Merge Sort
- (f) Which data structure is needed to convert infix notation to postfix notation?
- (a) Tree
  - (b) Graph
  - (c) Stack
  - (d) Queue
- (g) Evaluate the postfix expression  $ab + cd / -$  where  $a = 5, b = 4, c = 9, d = 3$ .
- (a) 23
  - (b) 15
  - (c) 6
  - (d) 10
- (h) The optimal data structure used to solve the Tower of Hanoi is \_\_\_\_\_.
- (a) Tree
  - (b) Heap
  - (c) Priority Queue
  - (d) Stack
- (i) What is a linear data structure with FIFO behavior? (Answer in one word).
- (j) What is the time complexity of Merge Sort? (Answer in one word).